Ontario Mathematics Competition

Monday, October 12, 2026 -
Friday, October 23, 2026

General Instructions:

1. **DO NOT** open the contest booklet until instructed by your proctor.
2. Before the contest begins, the proctors will give you a few minutes to read the instructions and fill in the contestant information section in your bubble sheet. There is no need to rush. Make sure to fill in the required fields legibly.
3. Diagrams are **NOT** drawn to scale. They are intended as aids only.
4. You are allowed scratch paper, a ruler, a compass, and a protractor for rough work.
5. Calculators are permitted as long as they do not have any of the following features:
   (i) internet access
   (ii) the ability to communicate with other devices
   (iii) information previously stored by students (such as formulas, programs, notes, etc.)
   (iv) a computer algebra system
   (v) dynamic geometry software
   Graphing calculators (GDCs) are **NOT** allowed.
6. Please do not discuss the contents of this exam online before 11:59 PM on October 23, 2026.

Exam Format:

1. The OMC consists of twenty-five multiple-choice questions to be completed in 60 minutes.
2. Each question is followed by answers marked A, B, C, D, and E. There is only one correct answer for each question.
3. Scoring: Each correct answer is worth 3 marks.
   There is no penalty for an incorrect answer.
   Each unanswered question is worth 1 mark.
   For tiebreaks, a tiebreaker score will be calculated where a correct answer is worth the same number of marks as its question number. For example, question 1 is worth 1 mark.

1. A notebook costs twice as much as a pen. A notebook and two pens cost \$12 altogether. How much does one notebook cost?

   A. \$2 B. \$3 C. \$4 D. \$5 E. \$6

2. The numbers $1, 2, 3, 4, 5, 6, 7, 8, 9$ are written on a board. Two numbers are erased, and the sum of the seven remaining numbers is 35. How many different pairs of numbers could have been erased?

   A. 4 B. 5 C. 8 D. 9 E. 10

3. A jar contains fewer than 30 beads. When the beads are arranged in groups of 4, there are 3 beads left over. When they are arranged in groups of 5, there are 4 beads left over. How many beads are in the jar?

   A. 9 B. 11 C. 14 D. 15 E. 19

4. What is the smallest possible radius of a circle that passes through the points $(0, 0)$ and $(1, 1)$?

   A. $\frac{1}{2}$ B. $\frac{\sqrt{2}}{2}$ C. $\frac{\sqrt{3}}{2}$ D. 1 E. 2

5. The graphs of $y = x^2 - x + 2$ and $y = x^2 + bx + c$ have at least two distinct points in common. What is the value of $b + c$?

   A. -3 B. -1 C. 1 D. 2 E. 3

6. Rose writes four distinct positive integers on a blackboard. Calum notices that their mean is 4. Calum then writes an additional number on the board, and Rose notes that the median of all five numbers is 6. What is the product of the original four numbers?

   A. 16 B. 42 C. 48 D. 80 E. 84

7. A positive integer is divisible by 36, and each digit in its decimal representation is either 0 or 1. What is the smallest possible number of digits it can have?

   A. 10 B. 11 C. 12 D. 18 E. 36

8. Each square of a 10×10 grid is labeled with the fewest moves needed to travel from that square to the top-left square, where each move goes to a square sharing a side. What is the sum of all the labels?

   A. 90 B. 450 C. 900 D. 1000 E. 1100

9. An 8×8 square grid has $9 \times 9 = 81$ lattice points, the vertices of its unit cells. Some of these points are colored red so that every unit cell has at least three red vertices. What is the smallest possible number of red points?

   A. 25 B. 48 C. 52 D. 56 E. 61

10. The largest positive divisor of a positive integer N, other than N itself, is 35. What is the greatest possible value of N?

    A. 35 B. 70 C. 105 D. 175 E. 245

11. Let ω be the unit circle centered at the origin. Initially, one point P is marked on ω. More points are then marked by repeatedly applying the following rules until no new points can be added:

    i. If a point on ω lies on the same vertical line as a marked point, mark it.
    ii. If a point on ω lies on the same horizontal line as a marked point, mark it.
    iii. If distinct points $A, B \in \omega$ satisfy that B is marked and line AB has slope 1, mark A.

    What is the sum of the maximum and minimum possible numbers of distinct marked points, over all choices of the initial point P?

    A. 6 B. 8 C. 10 D. 12 E. 16

12. Four or more distinct prime numbers are chosen. The product of each pair of chosen primes is calculated, and these products are added. What is the smallest possible sum that is divisible by 3?

    A. 21 B. 87 C. 141 D. 210 E. 213

13. There is a unique pair of positive integers (x, y) satisfying $x^2 - y^2 = 5x$. What is xy?

    A. 48 B. 54 C. 60 D. 72 E. 84

14. There are N distinct arrangements of the letters in MISSISSIPPI in which no two S's are adjacent. What is the sum of the digits of N?

    A. 5 B. 6 C. 7 D. 10 E. 15

15. What is the sum of the 199th powers of all 200 complex roots, counted with multiplicity, of

$$x^{200} + 15x - 5 = 0?$$

    A. -2985 B. -2970 C. -15 D. 0 E. 15

16. An integer n satisfies $2 \leq n \leq 1000$. Exactly eight of the fractions

$$\frac{1}{n}, \frac{2}{n}, \ldots, \frac{n-1}{n}$$

    have prime denominators when written in lowest terms. How many possible values of n are there?

    A. 6 B. 8 C. 9 D. 15 E. 16

17. In triangle ABC, $\angle B = 45^\circ$ and $\angle A = 5\angle C$. Given that $AC = 2\sqrt{2}$, what is $AB^2 + BC^2$?

    A. 8 B. 16 C. $8 + 8\sqrt{2}$ D. 32 E. 64

18. A rectangular grid consists of 4 rows and 12 columns of unit squares. A robotic ladybug travels from the bottom-left corner to the top-right corner, staying on the edges of these squares. Each move is one unit right, up, or down. The ladybug may never traverse the same unit edge twice.

    The ladybug chooses uniformly at random among all valid paths satisfying these conditions that reach the endpoint. Taking the bottom-left corner as the origin, let $P = (4, 2)$ and $Q = (8, 2)$. What is the probability that the ladybug's path passes through both P and Q?

    A. $\frac{289}{625}$ B. $\frac{17}{25}$ C. $\frac{441}{625}$ D. $\frac{561}{625}$ E. $\frac{593}{625}$

19. In triangle ABC, $AB = 14$ and $AC = 10$. Point D lies in the interior of side BC, with $AD = 10$ and $BD = 6$. Let O_1 and O_2 be the circumcenters of triangles ABD and ABC, respectively. The distance between O_1 and O_2 can be expressed as $m/\sqrt{n}$, where m, n are positive integers and n is not divisible by the square of any prime. What is mn?

    A. 14 B. 17 C. 21 D. 42 E. 84

20. For how many integers m with $1 \le m \le 100$ is there a nondegenerate triangle whose side lengths a, b, c satisfy

$$a^2 + b^2 + c^2 - \frac{1}{a+b+c} = ab + bc + ca$$

    and

$$\frac{a}{b+c-a} + \frac{b}{c+a-b} + \frac{c}{a+b-c} = m?$$

    A. 1 B. 96 C. 97 D. 98 E. 100

21. A machine initially displays 1. Each press of a button replaces the displayed integer n with

$$n + \gcd(n, 17) + \operatorname{lcm}(n, 17),$$

    where gcd and lcm denote the greatest common divisor and least common multiple, respectively. For example, the first press changes 1 to 19. How many presses are needed for the displayed number to be divisible by 289 for the first time?

    A. 16 B. 20 C. 23 D. 24 E. 32

22. In triangle ABC, $AB = 4$, $AC = 5$, and $BC = 6$. The internal angle bisectors of $\angle B$ and $\angle C$ meet AC and AB at D and E, respectively. Extend AB beyond B to F and AC beyond C to G so that $BF = BC = CG$.

    What is the ratio ED/FG?

    A. $\frac{4}{121}$ B. $\frac{2}{11}$ C. $\frac{1}{5}$ D. $\frac{2}{5}$ E. $\frac{5}{11}$

23. Arcane has five cats weighing 2, 4, 8, 16, and 32 kilograms. She places them on a seesaw one at a time, choosing a remaining cat and a side on each turn. After every placement, the total weight on the left side must exceed the total weight on the right side. Two sequences are considered different if the order of the cats or the side chosen for any cat differs. How many such sequences place all five cats on the seesaw?

    A. 945 B. 960 C. 1890 D. 1920 E. 3840

24. Consider the sum

$$S = \sum_{k=2}^{2027} k^2(k+1)!.$$

    What is the largest positive integer n such that $n!$ divides S?

    A. 2026 B. 2027 C. 2028 D. 2029 E. 2030

25. An arrow is drawn inside each square of a 2×6 rectangular grid of squares, pointing up, down, left, or right. A square is "pointed to" if an arrow in a square sharing a side with it points towards it. How many ways are there to draw the arrows so that every square is pointed to?

    A. 13 B. 121 C. 144 D. 156 E. 169

(THIS REMAINDER OF THIS PAGE IS INTENTIONALLY LEFT BLANK)
You may use this section for rough work. Only answers filled in the bubblesheet will be evaluated.
Partial marks are not awarded for work shown.

OMC

Student Name (Print Legibly) : _____
(FAMILY NAME) (GIVEN NAME)

School : _____

Grade : 9 ○ 10 ○ 11 ○ 12 ○ Other ○ : _____

Contestant ID						
	School ID			Test ID		
	⓪	⓪	⓪	⓪	⓪	⓪
	①	①	①	①	①	①
	②	②	②	②	②	②
	③	③	③	③	③	③
	④	④	④	④	④	④
	⑤	⑤	⑤	⑤	⑤	⑤
	⑥	⑥	⑥	⑥	⑥	⑥
	⑦	⑦	⑦	⑦	⑦	⑦
	⑧	⑧	⑧	⑧	⑧	⑧
	⑨	⑨	⑨	⑨	⑨	⑨

FILLING INSTRUCTIONS:

1. If you registered on our website, fill in your OCMC Contestant ID (Found as User ID on the website) in the space to the right.
   - ❖ You may safely ignore this section if you do not know your contest id.
2. Write your name, school and grade at the top.
3. Partial marks will not be awarded if your final answer is incorrect, even if work is shown. Answers outside the designated bubbles will not be marked.
4. Use a pencil to fill in the bubbles. To change an answer, erase completely before filling in a new choice.
5. To answer a question, completely blacken the corresponding circle. A circle is blackened if the white paper beneath is not visible. DO NOT fill multiple bubbles for the same question.
6. Do not use the margins as scratch paper.

Shading Methods

Incorrect Shading: ⊗ ⊘ ◉ ◐ ◑

Correct Shading: ○ ○ ○ ○ ●

Contest Answers

1 A ○ B ○ C ○ D ○ E ○	11 A ○ B ○ C ○ D ○ E ○	21 A ○ B ○ C ○ D ○ E ○
2 A ○ B ○ C ○ D ○ E ○	12 A ○ B ○ C ○ D ○ E ○	22 A ○ B ○ C ○ D ○ E ○
3 A ○ B ○ C ○ D ○ E ○	13 A ○ B ○ C ○ D ○ E ○	23 A ○ B ○ C ○ D ○ E ○
4 A ○ B ○ C ○ D ○ E ○	14 A ○ B ○ C ○ D ○ E ○	24 A ○ B ○ C ○ D ○ E ○
5 A ○ B ○ C ○ D ○ E ○	15 A ○ B ○ C ○ D ○ E ○	25 A ○ B ○ C ○ D ○ E ○
6 A ○ B ○ C ○ D ○ E ○	16 A ○ B ○ C ○ D ○ E ○	
7 A ○ B ○ C ○ D ○ E ○	17 A ○ B ○ C ○ D ○ E ○	
8 A ○ B ○ C ○ D ○ E ○	18 A ○ B ○ C ○ D ○ E ○	
9 A ○ B ○ C ○ D ○ E ○	19 A ○ B ○ C ○ D ○ E ○	
10 A ○ B ○ C ○ D ○ E ○	20 A ○ B ○ C ○ D ○ E ○	